



FOREST VERIFICATION REPORT

ANALYZED ZONE INFORMATION

Latitude	50.9983	Longitude	8.7908
Zone size (km)	10	Analysis date	2026-09-28

75

FOREST INTEGRITY SCORE / 100 Healthy Forest

DETAILED RESULTS

Indicator	Value	Interpretation
Current vegetation (NDVI)	0.749	Excellent
NDVI Trend 3 years (Sentinel-2)	+0.014	Stable
10-year trend (Landsat 8/9)	+0.438	Improving
40-year trend (Landsat 1984)	+0.456	Improving
Forest Status (FAO)	Forest recovery	Recuperation
Annual NDVI rate	+1.14% / year	Amelioration
NDVI change 5 years	+0.038	Stable
Leaf type	Mixed (Coniferous/Broadleaf)	Mixed needles and flat leaves
Developmental stage	Mature forest (60-120 years)	Well established, dense canopy
Above-ground biomass	~40.6 t/ha	Low biomass (<50 t/ha)
Canopy Height (GEDI+S2 10m)	9.5m max / 3.5m median	Tree height from LiDAR (ISS) fused with Sentinel-2 at 10m resolution
Carbon Assessment (IPCC)	694,294 t CO2	CO2 sequestered in this forest zone — IPCC methodology (AGB+BGB × 0.47 × 3.67)
CO2 per hectare	88.4 t CO2/ha	Carbon density per hectare
Zone area	7,854 ha	Analysis zone area in hectares
Carbon credit value	EUR 10,414,410	Estimated value at EUR 15/tonne CO2 (voluntary carbon market)
REDD+ Compatibility	REDD+ Compatible	Satellite monitoring data compatible with REDD+ MRV requirements (VCS VM0007, VM0009). Independent verification by accredited auditor required for official credit issuance.
Methodology reference	IPCC 2006/2019	IPCC Guidelines for National Greenhouse Gas Inventories (2006, 2019 refinement) — Tier 1 approach. Zenodo DOI: 10.5281/zenodo.22872779

GEDI Tree Cover	52.1%	Percentage of the zone covered by trees according to NASA GEDI
AI Forest Type	Dense Tropical Forest	High biodiversity, dense canopy
Sentinel-1 Radar (VV)	-9.698 dB	Dense forest (radar)
Forest cover	99.1%	Excellent
Fire pixels detected (FIRMS/year)	238 pixels	~60 events estimated
Overall trend	stable	
Satellites used	- ESA Sentinel-2 - NASA FIRMS - ESA Sentinel-1 - NASA Landsat 8/9 - NASA Landsat 4/5	
Deforestation Alert	STABLE — No significant change	
Area change (3 years)	+14.4 ha	+0.09%
Annual rate (area/yr)	+4.8 ha/yr	Recovery
Area change (10 years)	+14946.1 ha	+2013.88%
Annual rate 10yr (area)	+1494.6 ha/yr	Recovery

SCORE BREAKDOWN

Component	Points	Maximum
Current vegetation (NDVI)	25	25
NDVI trend (3 years)	15	25
Forest cover	25	25
Sentinel-1 Radar (VV)	10	15
Fire score (0=many fires, 10=none)	0	10
TOTAL	75	100

FIRE DETECTION METHODOLOGY: Fire pixels are detected by NASA FIRMS using VIIRS (375m) and MODIS (1km) thermal anomaly sensors. Raw pixel count represents thermal detections over 12 months in the analysis zone. Estimated fire events are calculated using NASA clustering methodology (~4 pixels per fire event average). This is a statistical approximation. For precise fire event mapping, consult NASA FIRMS directly at firms.modaps.eosdis.nasa.gov

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